

Zenodo Biceps sEMG Dataset

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Dataset Understanding Guide — ENSE891 Final Year Project

Subjects:	13
Sample rate (EMG):	1259 Hz (Delsys)
Label rate:	50 Hz (self-reported)
Muscles per trial:	4 (bicep + 3 deltoids)
Trials per subject:	12 (6 R arm, 6 L arm)
Window (authors' recipe):	4s: 2s step
Total windows (trial 5 + 6):	3300 across 13 subjects
Labels:	0 = Fresh 1 = Transition 2 = Fatigued
Subject 6 note:	Only 24.5s trial 5 — excluded from classifier
MVC files:	Per subject, per muscle — max voluntary contraction baseline

Dataset File Structure

```
zenodo_biceps/
  sEMG_data/
    subject_1/ ... subject_13/
      trial_1.csv ... trial_12.csv      <- EMG signals at 1259 Hz
    MVC/
      subject_N MVC_R_BICEPS.csv      <- max voluntary contraction
      subject_N MVC_L_BICEPS.csv
      subject_N MVC_R_DELTAOID_ANTERIOR.csv
      ... (8 MVC files per subject)
  self_perceived_fatigue_index/
    subject_1/ ... subject_13/
      Trial_1.csv ... Trial_12.csv      <- fatigue labels at 50 Hz

EMG trial CSV columns (odd-numbered trials = Right side):
X [s]                                time
R BICEPS BRACHII: EMG 1 [V]          primary muscle for THIS project
X [s].1                              (duplicate time col for next muscle)
R DELTOID ANTERIOR: EMG 2 [V]
X [s].5
R DELTOID MEDIUS: EMG 6 [V]
X [s].6
R DELTOID POSTERIOR: EMG 7 [V]

Label CSV columns:
time    seconds since trial start (50 Hz)
label   0 = Fresh   |  1 = Transition |  2 = Fatigued

MVC CSV columns:
X [s]                                time
R BICEPS BRACHII: EMG 1 [V]          raw signal
R BICEPS BRACHII: EMG 1->filter [V]  sliding RMS filtered
```

Key Notes:

- Each trial has exactly 4 muscles recorded simultaneously.
- Odd trials (1,3,5,7,9,11) = RIGHT arm. Even trials (2,4,6,8,10,12) = LEFT arm.
- This project uses TRIAL 5 (R bicep) and TRIAL 6 (L bicep).
- The bicep column is at different positions: R trials = col 2, L trials = col 4.
- EMG amplitude is in Volts (millivolt range: typically 0.05-0.4 mV RMS).
- Label file rows are ~25x fewer than EMG rows (50 Hz vs 1259 Hz).
- MVC = Maximum Voluntary Contraction, used to normalise %MVC if needed.
- Subject 6 trial 5 is only 24.5s — too short for meaningful windowing, excluded.

Trial Map — All 12 Trials

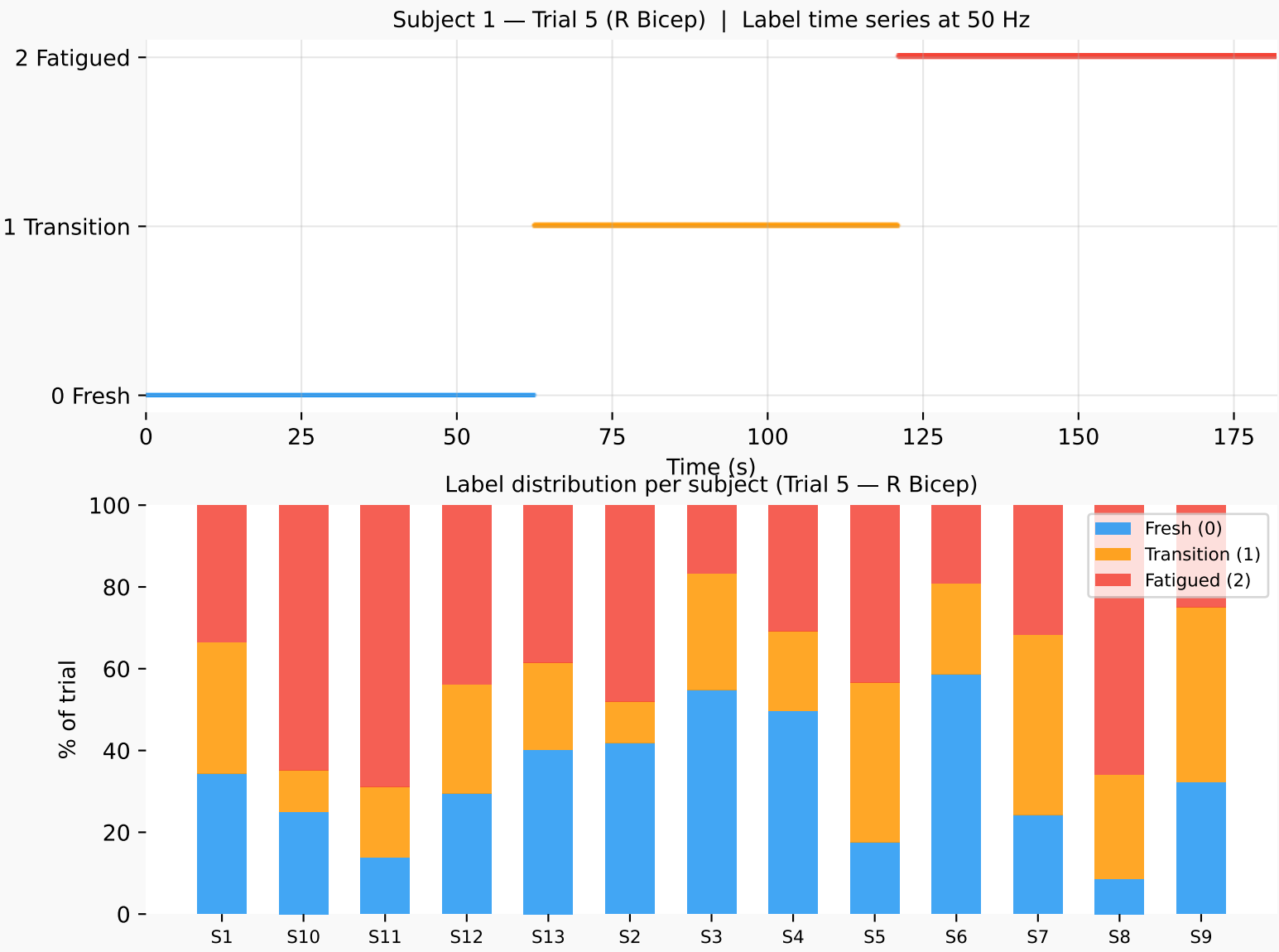
Each trial records 4 muscles. Trials alternate Right / Left arm.

Trial	Side	Muscles recorded	Used in project
1	R	R BICEPS R DELTOID_ANT R DELTOID_MED R DELTOID_POST	
2	L	L DELTOID_ANT L BICEPS L DELTOID_POST L DELTOID_MED	
3	R	R BICEPS R DELTOID_ANT R DELTOID_MED R DELTOID_POST	
4	L	L DELTOID_ANT L BICEPS L DELTOID_POST L DELTOID_MED	
5	R	R BICEPS R DELTOID_ANT R DELTOID_MED R DELTOID_POST	Yes (Bicep only)
6	L	L DELTOID_ANT L BICEPS L DELTOID_POST L DELTOID_MED	Yes (Bicep only)
7	R	R BICEPS R DELTOID_ANT R DELTOID_MED R DELTOID_POST	
8	L	L DELTOID_ANT L BICEPS L DELTOID_POST L DELTOID_MED	
9	R	R BICEPS R DELTOID_ANT R DELTOID_MED R DELTOID_POST	
10	L	L DELTOID_ANT L BICEPS L DELTOID_POST L DELTOID_MED	
11	R	R BICEPS R DELTOID_ANT R DELTOID_MED R DELTOID_POST	
12	L	L DELTOID_ANT L BICEPS L DELTOID_POST L DELTOID_MED	

Why only trials 5 and 6?

Trials 1-4 and 7-12 record from deltoid muscles (shoulder).
The project targets BICEP BRACHII fatigue classification.
Trials 5 (R bicep) and 6 (L bicep) are the fatigue-to-exhaustion
dumbbell curl trials — the primary experimental condition.
Deltoid trials serve as synergist context but are not the fatigue target.
The 12-trial design lets researchers compare fatigue across muscle groups;
we use only the bicep subset that matches our research question.

Label Format and Fatigue Protocol



How the labels work

Labels are self-perceived fatigue, recorded continuously during the bicep curl protocol. The subject performs dumbbell curls to exhaustion while pressing a button scale to report their fatigue.

- Label 0 – Fresh: subject feels no fatigue. Occurs at start of trial.
- Label 1 – Transition: subject is beginning to feel fatigue (mid-trial).
- Label 2 – Fatigued: subject reports significant fatigue (end of trial).

Rate: 50 Hz (one label every 0.02s). The EMG runs at 1259 Hz, so ~25 EMG samples per label sample.

Alignment: for windowed classification (4s windows), the label for a window is taken as the MODE (most frequent label) within that window's time range in the label file.

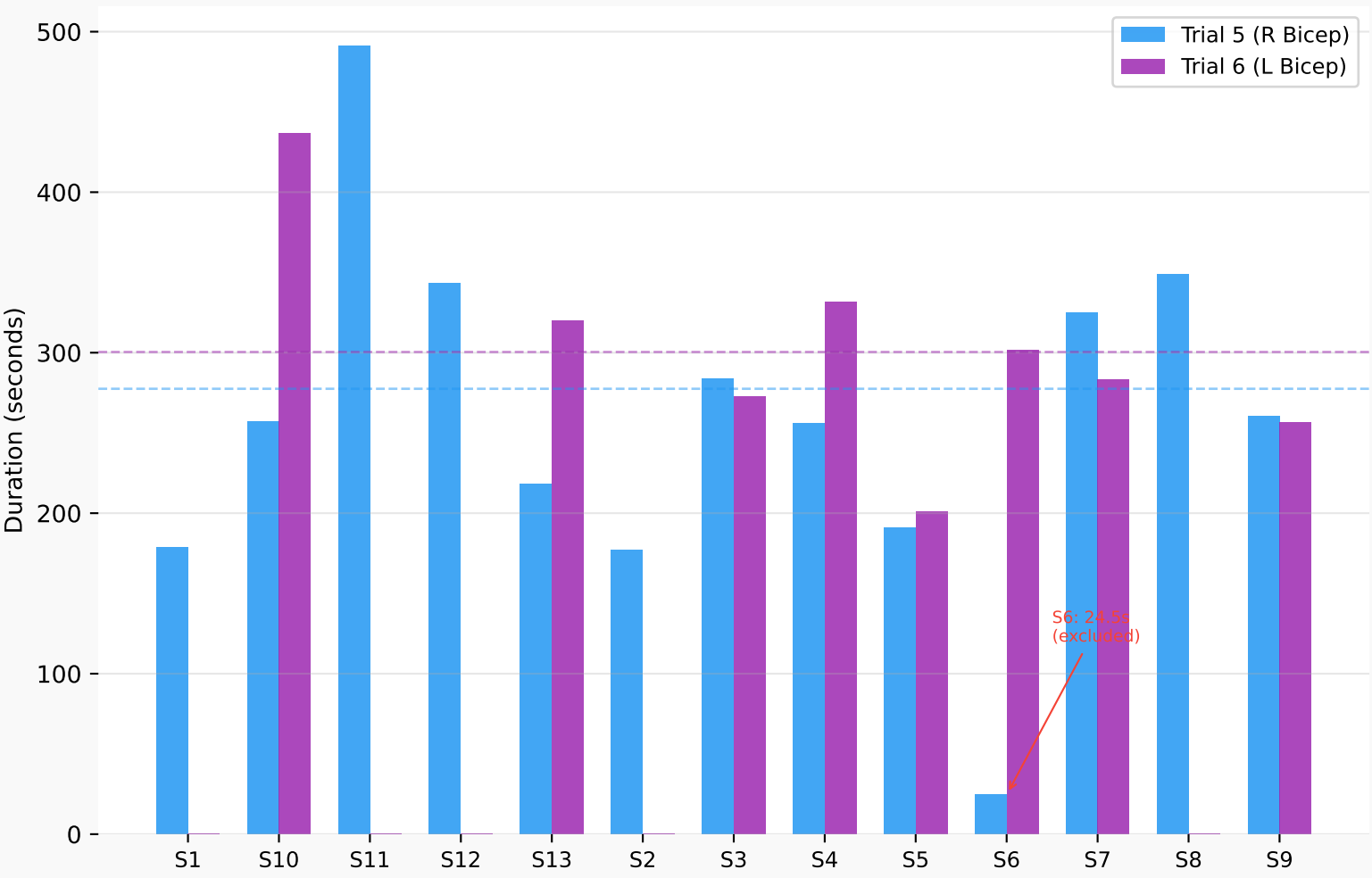
Why is accuracy hard on class 1?

Label 1 (Transition) is inherently ambiguous – the subject themselves are unsure. The EMG signal may not yet show clear fatigue markers at this point. This explains why 3-class classification sits at ~68% even for SVM, while binary (0 vs 2) reaches 88%.

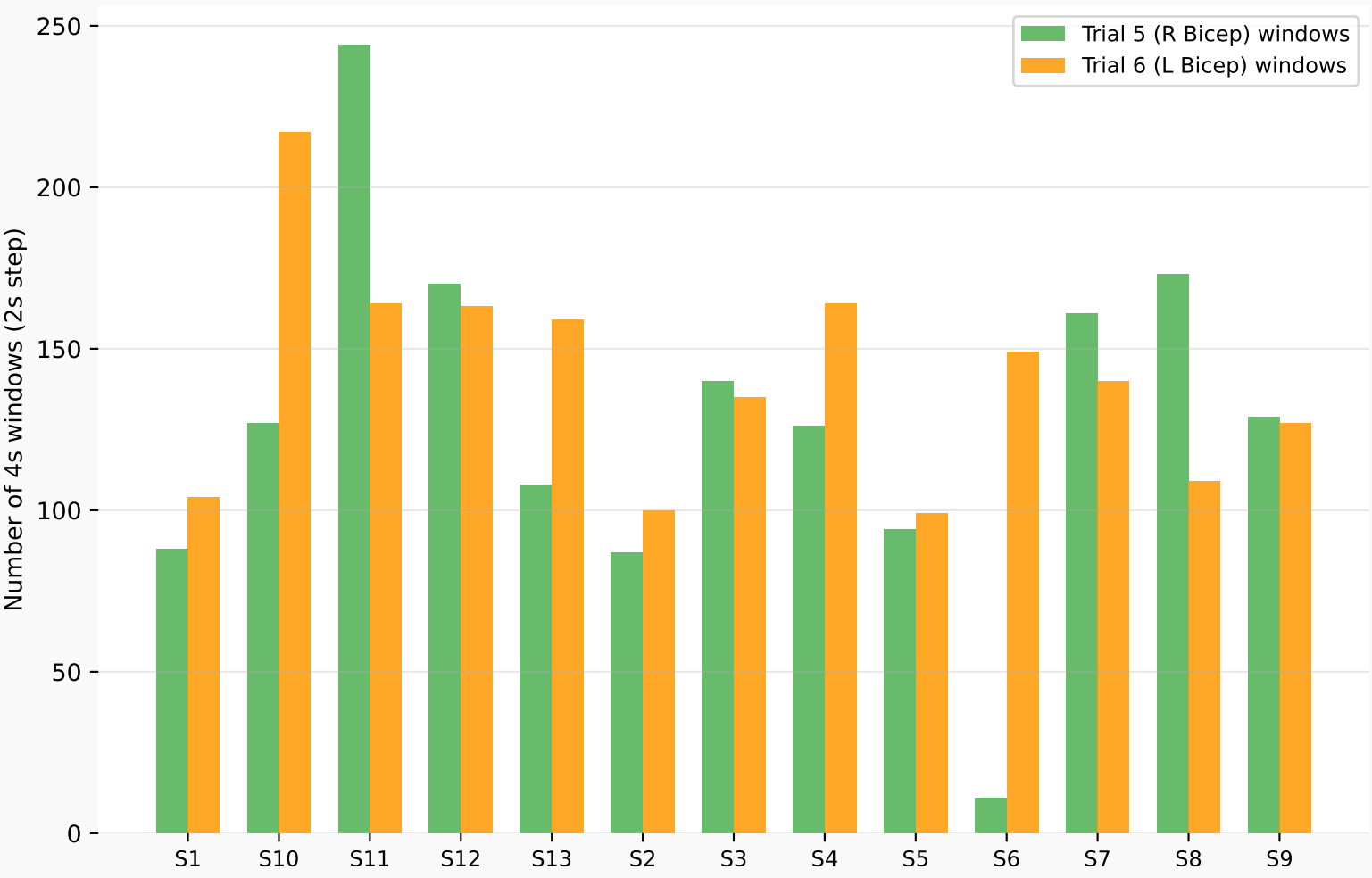
Important: labels vary enormously between subjects. S3 spends 55% of trial in class 0 (slow fatiguer). S8 spends 71% in class 2 (fast fatiguer). This inter-subject variation is exactly why cross-subject (LOSO) accuracy is lower than within-subject accuracy.

Per-Subject Statistics

Trial duration per subject (dashed = mean, excluding S6 short trial)

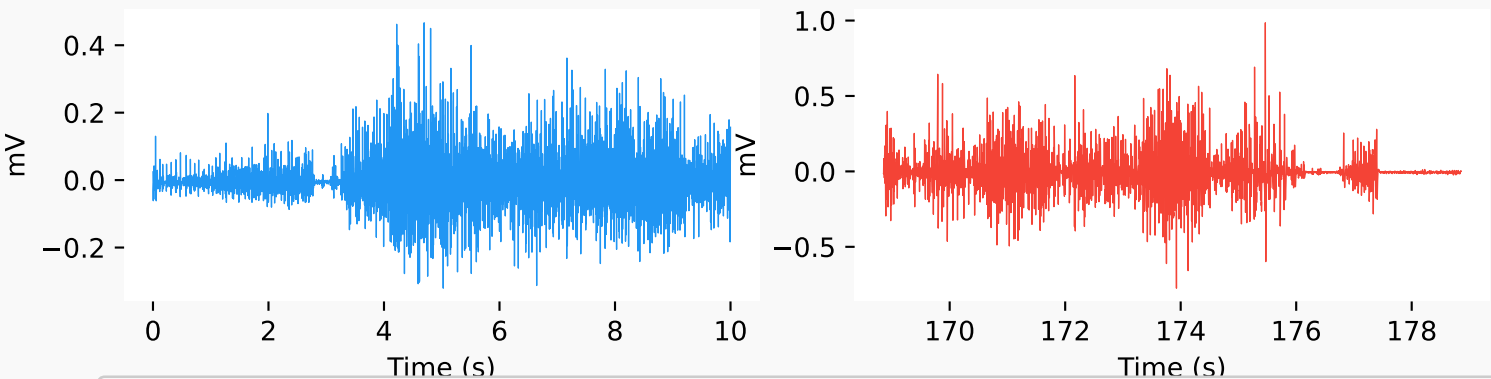
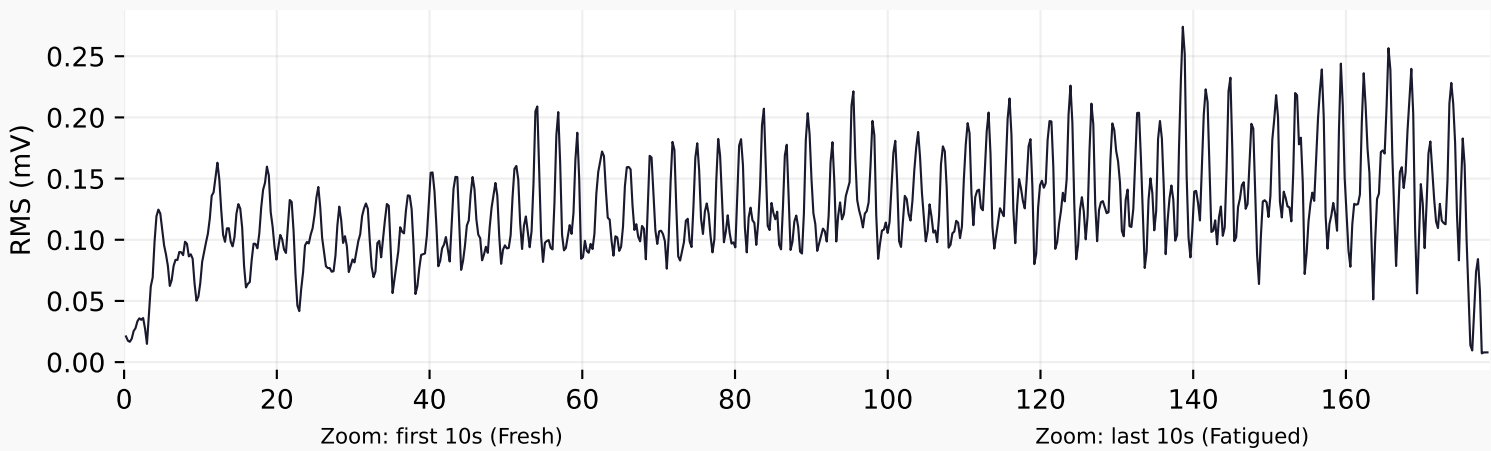
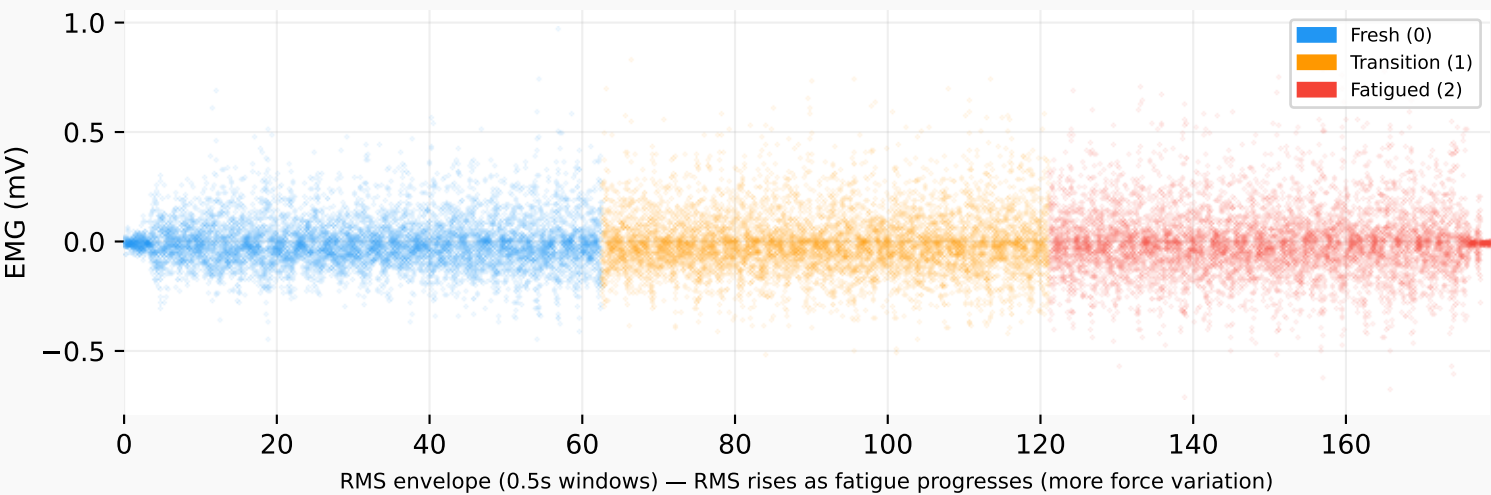


Windows available per subject | Total both arms: 3488



EMG Signal Walkthrough — Subject 1, Trial 5 (R Bicep)

Raw EMG signal coloured by label (Blue=Fresh, Orange=Transition, Red=Fatigued)



Reading the EMG signal

Amplitude: raw values in Volts, millivolt range (typ. 0.05-0.4 mV RMS for bicep curls).
Shape: AC signal oscillating around zero. Each burst = one muscle contraction (one curl rep).
The gaps between bursts = rest between reps.

Fatigue markers visible in the signal:

- Amplitude: tends to INCREASE as fatigue progresses (more motor units recruited to maintain force).
- Frequency content: shifts DOWNWARD (median frequency drops) as fatigue progresses. This is the MDF-decline effect shown in the slides.
- Burst shape: becomes more irregular and prolonged near exhaustion.

What classifiers learn:

RF/SVM/KNN: manually extracted features (RMS, MDF, zero-crossing rate per window).
Transformer: feeds raw windowed signal and learns features itself via attention.
It can discover frequency-time patterns that hand-crafted features miss.

Sample rate note: 1259 Hz is higher than needed for EMG (Nyquist: 2x max freq ~500 Hz).
Delsys hardware defaults to this rate. The authors' recipe bandpasses 20-450 Hz.
For the transformer, we can either use raw 1259 Hz or downsample to 250 Hz to match the OpenBCI rig for future real-world deployment.

Key Notes for This Project

Dataset source

Zenodo 10.5281/zenodo.14182446, Sensors 2024 paper. CC BY 4.0. Cite it.

Why bicep?

No clean forearm public dataset exists. Bicep curl to exhaustion gives a clean, labelled fatigue trajectory. Protocol matches our planned self-collection (age bands).

Subject exclusion

Subject 6 trial 5 = only 24.5s. Results in only 11 windows — too few for LOSO. Excluded from all classifiers. 12 subjects used.

--norm flag

Normalise each window's features by that subject's own resting baseline. Critical for 3-class: takes LOSO from 39% to 67%. Always use --norm for cross-subject work.

3-class vs binary

Binary (0 vs 2) = 88% accuracy RF. 3-class = 68% SVM. Transition class is noisy. Binary result is publishable; 3-class is the harder academic contribution.

Transformer next step

RF/SVM/KNN results become the baseline. Transformer (PatchTST style) feeds raw 4s windows. Target: beat 68% 3-class. Even 72% is a publishable improvement.

Age bands (future)

Zenodo does not include age metadata. Age-band analysis requires self-collected data (bicep curls, same protocol, 4 age groups). Dataset and pipeline are proven; recording is the remaining step.

Window alignment

Authors: 4s window, 2s step = 50% overlap. Window label = mode of label samples within that window. Windows from fresh/fatigued ends are cleanest. Transition windows are noisy.

MVC files

Contain max voluntary contraction recording. Used to compute %MVC normalisation if needed. Not currently used in classifier — could be used as a per-subject normalisation baseline instead of --norm.

1259 Hz vs 250 Hz

Delsys records at 1259 Hz. OpenBCI records at 250 Hz. If we train at 1259 Hz and deploy on OpenBCI, we must resample. Consider downsampling to 250 Hz before feature extraction to bridge the gap.

Left vs Right arm

Trials 5 (R) and 6 (L) have different durations per subject. Both arms show fatigue but timing differs. Using both doubles window count (~3300 total) which helps transformer training.

Classifier results

RF: 67.1%, SVM: 68.2%, KNN: 64.1% (3-class LOSO --norm). These are the numbers the transformer must beat.